

John Carter Lynch

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SUMMARY

Software developer with 2 YOE building backend systems and applications. Focused on Go/C++, with expertise in REST APIs, database architectures, concurrency, computer networking, Agile, and testing methods. Brings practical defense-sector knowledge of SIGINT, EW, and wireless communications.

EDUCATION

2019 - 2023 Computer Science at **UC Santa Cruz**

GPA: 3.7

SKILLS

Languages:	Go, C++, Python, SQL, JavaScript, Bash
Frameworks & Libraries:	ProtoBuf, gRPC, SQLite, Next.js, React
Developer Tools:	Git, Linux, OpenWRT, Wireshark, Docker, AWS, PostgreSQL

EXPERIENCE

BlueHalo | Software Developer - SIGINT & EW March 2024 - February 2025

- Developed backend in Go for a signals intelligence and electronic warfare system, processing signal scan results from embedded, orchestrating cyber operations, and streaming mission data to the UI.
- Key role in successfully adding platform support for Bluetooth LE, extending backend services and ProtoBuf API, allowing operators to detect and exploit 10× more Bluetooth devices in the field.
- Rebuilt the audit-logging system to a generic and event-driven design, adding support for new operation types and eliminating log-related issues.
- Wrote headless gateway simulators, unit tests, and end-to-end test suites, allowing developers and QA to validate features without RF hardware and shorten manual test cycles.
- Partnered with frontend, embedded, and QA teams to identify and resolve critical issues using Wireshark and debugging tools, delivering fixes ahead of field deployments.
- Peer-reviewed operator guides and authored technical documentation, improving maintainability and shortening onboarding time for new engineers.

Prometheus Consortium | Software Developer (Contract) January 2023 - January 2024

- Integrated new features into blockchain infrastructure written in Go, collaborating with startup founders to successfully build a proof-of-concept blockchain product that aligned with the company's strategy.

GoGoPool | Software Developer Intern May 2022 - August 2022

- Built web application in JavaScript and backend API in Go for the GoGoPool De-Fi protocol, enabling interaction with on-chain smart contracts and gaining experience in crafting user-facing products.
- Led development of internal monitoring tools, including a web application and widget, improving the team's tracking and monitoring capabilities.

PROJECTS

VAFilterSynth [Github link](#)

- Built a digital synth in C++, implementing DSP concepts such as PolyBLEP and oversampling, and featuring a nonlinear digital emulation of an analog diode ladder voltage controlled filter.

Tiny-Async-HTTP [Github link](#)

- Created a scalable and asynchronous HTTP server library in Rust, mirroring the simplicity of Express.js while supporting high-volume concurrent connections.